

# Relation

A software tool for exploring the relation between diagnostic accuracy and measurement uncertainty

Interface Documentation

Version 1.0.6

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## 1. About Program Controls

The numerical settings are defined by the user with menus or sliders. Sliders can be finely manipulated by holding down the *alt* key or *opt* key while dragging the mouse. They can be even more finely manipulated by also holding the *shift* and/or *ctrl* keys.

Dragging with the mouse rotates the three-dimensional plots, while dragging with the mouse while pressing the *ctrl*, *alt*, or *opt* keys zooms in or out.

## 2. Range of the Input Parameters

$v$ : 0.00000001 – 0.99

$\mu_D, \mu_{\bar{D}}$ : 0 - 6

$\sigma_D, \sigma_{\bar{D}}$ : 0.01 - 6

$d$ : Minimum of  $(\mu_D - 3.5\sigma_D, \mu_{\bar{D}} - 3.5\sigma_{\bar{D}})$  – Maximum of  $(\mu_D + 3.5\sigma_D, \mu_{\bar{D}} + 3.5\sigma_{\bar{D}})$

$u$ : 0 - 6

$l_0$ : 0 - 1000

$l_{TN}$ : 0 - 1000

$l_{FN}$ : 0 - 1000

$l_{TP}$ : 0 - 1000

$l_{FP}$ : 0 - 1000

Initial plot points per axis: 10 – 50

### 3. Range of the Coordinates of the Plots

There are two options for the range of coordinates to be included in each plot:

- 1) *Full*: All the calculated coordinate points are plotted.
- 2) *Partial*: The distribution of coordinate values is found and any points sufficiently far out in the distribution are not considered.

### 4. Input and Output

The program provides in five modules and six submodules plots and tables of diagnostic accuracy measures of two screening or diagnostic tests for a single measurand, differing in measurement uncertainty, applied at a single point in time in a diseased and a nondiseased population.

Singularity points are excluded from the plots.

Indeterminate results of the calculation modules arise when numerical quantities cannot be resolved because they are outside the computational range.

#### 4.1. ROC curves

Input

The user defines:

- 1) The prevalence of the disease.
- 2) The means and the standard deviations of the measurand in the diseased and nondiseased populations.
- 3) The standard measurement uncertainty of each test.

Output

- 1) a) The plots of the *ROC* curves of the two tests and  
b) A table with the areas under and over the *ROC* (Receiver Operating Characteristic) curves (*AUC* and *AOC* respectively), or
- 2) The plots of the *PROC* (Predictive Receiver Operating Characteristic) curves of the two tests.

#### 4.2. Diagnostic accuracy measures plots

The module presents in three submodules plots of diagnostic accuracy measures of the tests.

##### 4.2.1. Diagnostic accuracy measures versus diagnostic threshold

Input

- 1) The user defines:
  - a) The prevalence of the disease.
  - b) The means and the standard deviations of the measurand in the diseased and nondiseased populations.
  - c) The standard measurement uncertainty of each test.
  - d) The expected loss for:
    - i) Testing,
    - ii) A true positive result,
    - iii) A true negative result,
    - iv) A false positive result and

- v) A false negative result.
- 2) The diagnostic accuracy measure function to be plotted, by selecting:
  - a) The measure:
    - i) Sensitivity ( $Se$ ),
    - ii) Specificity ( $Sp$ ),
    - iii) Overall diagnostic accuracy ( $ODA$ ),
    - iv) Positive predictive value ( $PPV$ ),
    - v) Negative predictive value ( $NPV$ ),
    - vi) Diagnostic odds ratio ( $DOR$ ),
    - vii) Likelihood ratio for a positive test result ( $LR+$ ),
    - viii) Likelihood ratio for a negative test result ( $LR-$ ),
    - ix) Youden's index ( $J$ ),
    - x) Euclidean distance ( $ED$ ),
    - xi) Concordance probability ( $CZ$ ),
    - xii) Risk ( $R$ ).
  - b) The codomain of the function, that is
    - i) The value of the measure,
    - ii) The partial derivative of the measure with respect to standard measurement uncertainty,
    - iii) The difference between the measures of the two tests,
    - iv) The relative difference between the measures of the two tests,
    - v) The ratio of the measures of the two tests.
  - c) The range of the coordinate points to be plotted:
    - i) Full,
    - ii) Partial.

## Output

Plots of:

- 1) The values of the measure of each test,
- 2) The partial derivatives of the measure of each test with respect to standard measurement uncertainty,
- 3) The difference between the measures of the two tests,
- 4) The relative difference between the measures of the two tests,
- 5) The ratio of the measures of the two tests,

versus diagnostic threshold ( $d$ ).

### 4.2.2. Diagnostic accuracy measures versus prevalence

## Input

The user defines:

- 1) The means and the standard deviations of the measurand in the diseased and nondiseased populations.
- 2) The diagnostic threshold.
- 3) The standard measurement uncertainty of each test.
- 4) The expected loss for:
  - a) Testing,
  - b) A true positive result,
  - c) A true negative result,
  - d) A false positive result and
  - e) A false negative result.
- 5) The diagnostic accuracy measure function to be plotted, by selecting:
  - a) The measure:
    - i) Sensitivity ( $Se$ ),
    - ii) Specificity ( $Sp$ ),
    - iii) Overall diagnostic accuracy ( $ODA$ ),
    - iv) Positive predictive value ( $PPV$ ),

- v) Negative predictive value ( $NPV$ ),
  - vi) Diagnostic odds ratio ( $DOR$ ),
  - vii) Likelihood ratio for a positive test result ( $LR+$ ),
  - viii) Likelihood ratio for a negative test result ( $LR-$ ),
  - ix) Youden's index ( $J$ ),
  - x) Euclidean distance ( $ED$ ),
  - xi) Concordance probability ( $CZ$ ),
  - xii) Risk ( $R$ ).
- b) The codomain of the function, that is
- i) The value of the measure,
  - ii) The partial derivative of the measure with respect to standard measurement uncertainty,
  - iii) The difference between the measures of the two tests,
  - iv) The relative difference between the measures of the two tests,
  - v) The ratio of the measures of the two tests.
- 6) The range of the coordinate points to be plotted:
- a) Full,
  - b) Partial.

#### Output

Plots of:

- 1) The values of the measure of each test,
- 2) The partial derivatives of the measure of each test with respect to standard measurement uncertainty,
- 3) The difference between the measures of the two tests,
- 4) The relative difference between the measures of the two tests,
- 5) The ratio of the measures of the two tests,

versus prevalence ( $v$ ).

#### 4.2.3. Diagnostic accuracy measures versus uncertainty

##### Input

The user defines:

- 1) The prevalence of the disease.
- 2) The means and the standard deviations of the measurand in the diseased and nondiseased populations.
- 3) The diagnostic threshold.
- 4) The expected loss for:
  - i) Testing,
  - ii) A true positive result,
  - iii) A true negative result,
  - iv) A false positive result and
  - v) A false negative result.
- 5) The diagnostic accuracy measure function to be plotted:
  - a) Sensitivity ( $Se$ ),
  - b) Specificity ( $Sp$ ),
  - c) Overall diagnostic accuracy ( $ODA$ ),
  - d) Positive predictive value ( $PPV$ ),
  - e) Negative predictive value ( $NPV$ ),
  - f) Diagnostic odds ratio ( $DOR$ ),
  - g) Likelihood ratio for a positive test result ( $LR+$ ),
  - h) Likelihood ratio for a negative test result ( $LR-$ ),
  - i) Youden's index ( $J$ ),
  - j) Euclidean distance ( $ED$ ),
  - k) Concordance probability ( $CZ$ ),

- l) Risk ( $R$ ).
- 6) The range of the coordinate points to be plotted:
  - i) Full,
  - ii) Partial.

#### Output

Plots of the values of the measure of each test versus uncertainty ( $u$ ).

### 4.3. Diagnostic accuracy measures relations plots

The module presents in three submodules plots of the relations between diagnostic accuracy measures of the tests.

#### 4.3.1. Diagnostic accuracy measures versus sensitivity or specificity

#### Input

The user defines:

- 1) The prevalence of the disease
- 2) The means and the standard deviations of the measurand in the diseased and nondiseased populations.
- 3) The standard measurement uncertainty of each test.
- 4) The expected loss for:
  - a) Testing,
  - b) A true positive result,
  - c) A true negative result,
  - d) A false positive result and
  - e) A false negative result.
- 2) The diagnostic accuracy measure function to be plotted by selecting:
  - a) The measure:
    - i) Overall diagnostic accuracy ( $ODA$ ),
    - ii) Positive predictive value ( $PPV$ ),
    - iii) Negative predictive value ( $NPV$ ),
    - iv) Diagnostic odds ratio ( $DOR$ ),
    - v) Likelihood ratio for a positive test result ( $LR+$ ),
    - vi) Likelihood ratio for a negative test result ( $LR-$ ),
    - vii) Youden's index ( $J$ ),
    - viii) Euclidean distance ( $ED$ ),
    - ix) Concordance probability ( $CZ$ ),
    - x) Risk ( $R$ ).
  - b) The domain of the function:
    - i) The sensitivity ( $Se$ ) of the test,
    - ii) The specificity ( $Sp$ ) of the test.
  - c) The codomain of the function, that is
    - i) The value of the measure,
    - ii) The partial derivative of the measure with respect to standard measurement uncertainty,
    - iii) The difference between the measures of the two tests,
    - iv) The relative difference between the measures of the two tests,
    - v) The ratio of the measures of the two tests.
- 3) The range of the coordinate points to be plotted:
  - a) Full,
  - b) Partial.

#### Output

Plots of:

- 1) The values of the measure of each test,
- 2) The partial derivatives of the measure of each test with respect to standard measurement uncertainty,

- 3) The difference between the measures of the two tests,
- 4) The relative difference between the measures of the two tests,
- 5) The ratio of the measures of the two tests,  
versus sensitivity ( $Se$ ) or specificity ( $Sp$ ).

#### 4.3.2. Diagnostic accuracy measures versus sensitivity and specificity

##### Input

The user defines:

- 1) The prevalence of the disease.
- 2) The means and the standard deviations of the measurand in the diseased and nondiseased populations.
- 3) The standard measurement uncertainty of each test.
- 4) The expected loss for:
  - a) Testing,
  - b) A true positive result,
  - c) A true negative result,
  - d) A false positive result and
  - e) A false negative result.
- 5) The diagnostic accuracy measure function to be plotted by selecting:
  - a) The measure:
    - i) Overall diagnostic accuracy ( $ODA$ ),
    - ii) Positive predictive value ( $PPV$ ),
    - iii) Negative predictive value ( $NPV$ ),
    - iv) Diagnostic odds ratio ( $DOR$ ),
    - v) Likelihood ratio for a positive test result ( $LR+$ ),
    - vi) Likelihood ratio for a negative test result ( $LR-$ ),
    - vii) Youden's index ( $J$ ),
    - viii) Euclidean distance ( $ED$ ),
    - ix) Concordance probability ( $CZ$ ),
    - x) Risk ( $R$ ).
  - b) The codomain of the function, that is
    - i) The value of the measure,
    - ii) The partial derivative of the measure with respect to standard measurement uncertainty.
- 6) The range of the coordinate points to be plotted:
  - a) Full,
  - b) Partial.

##### Output

Three-dimensional line plots of the values or the derivatives of the measure of each test versus its sensitivity ( $Se$ ) and specificity ( $Sp$ ).

#### 4.3.3. Diagnostic accuracy measures relations

##### Input

The user defines:

- 1) The prevalence of the disease.
- 2) The means and the standard deviations of the measurand in the diseased and nondiseased populations.
- 3) The standard measurement uncertainty of each test.
- 4) The expected loss for:
  - a) Testing,
  - b) A true positive result,
  - c) A true negative result,
  - d) A false positive result and
  - e) A false negative result.
- 5) The relation between the diagnostic accuracy measures to be plotted, by selecting any two of the following:

- a) Overall diagnostic accuracy (*ODA* ),
  - b) Positive predictive value (*PPV* ),
  - c) Negative predictive value (*NPV* ),
  - d) Diagnostic odds ratio (*DOR* ),
  - e) Likelihood ratio for a positive test result (*LR+*),
  - f) Likelihood ratio for a negative test result (*LR-*),
  - g) Youden's index (*J* ),
  - h) Euclidean distance (*ED* ),
  - i) Concordance probability (*CZ* ),
  - j) Risk (*R* ).
- 6) The range of the coordinate points to be plotted:
- a) Full,
  - b) Partial.

#### Output

Plots of the respective parametric equations of any two of the above diagnostic accuracy measures.

### 4.4. Calculators

#### 4.4.1. Diagnostic accuracy measures calculator

##### Input

The user defines:

- 1) The prevalence of the disease.
- 2) The means and the standard deviations of the measurand in the diseased and nondiseased populations.
- 3) The diagnostic threshold.
- 4) The standard measurement uncertainty of each test.
- 5) The expected loss for:
  - a) Testing,
  - b) A true positive result,
  - c) A true negative result,
  - d) A false positive result and
  - e) A false negative result,.

##### Output

A table of the values and the relative differences of the following diagnostic accuracy measures of the two tests for the selected diagnostic threshold:

- a) Sensitivity (*Se* ),
- b) Specificity (*Sp* ),
- c) Overall diagnostic accuracy (*ODA* ),
- d) Positive predictive value (*PPV* ),
- e) Negative predictive value (*NPV* ),
- f) Diagnostic odds ratio (*DOR* ),
- g) Likelihood ratio for a positive test result (*LR+*),
- h) Likelihood ratio for a negative test result (*LR-*),
- i) Youden's index (*J* ),
- j) Euclidean distance (*ED* ),
- k) Concordance probability (*CZ* ),
- l) Risk (*R* ).

#### 4.4.2. Optimal diagnostic accuracy measures calculator

##### Input

The user defines:

- 1) The prevalence of the disease.

- 2) The means and the standard deviations of the measurand in the diseased and nondiseased populations.
- 3) The diagnostic threshold.
- 4) The expected loss for:
  - a) Testing,
  - b) A true positive result,
  - c) A true negative result,
  - d) A false positive result and
  - e) A false negative result.
- 5) The objective or loss function to find the diagnostic threshold for the following optima:
  - a) Maximum Youden's index ( $J$ ),
  - b) Minimum Euclidean distance ( $ED$ ),
  - c) Maximum concordance probability ( $CZ$ ),
  - d) Minimum risk ( $R$ ).

Output

- 1) The optimal diagnostic thresholds for the two tests.
- 2) A table of the values of the following diagnostic accuracy measures of the two tests at the respective optimal diagnostic thresholds and of their relative differences:
  - a) Sensitivity ( $Se$ ),
  - b) Specificity ( $Sp$ ),
  - c) Overall diagnostic accuracy ( $ODA$ ),
  - d) Positive predictive value ( $PPV$ ),
  - e) Negative predictive value ( $NPV$ ),
  - f) Diagnostic odds ratio ( $DOR$ ),
  - g) Likelihood ratio for a positive test result ( $LR+$ ),
  - h) Likelihood ratio for a negative test result ( $LR-$ ),
  - i) Youden's index ( $J$ ),
  - j) Euclidean distance ( $ED$ ),
  - k) Concordance probability ( $CZ$ ),
  - l) Risk ( $R$ ).

## 5. About the Program

It is assumed that if a test measurement is above the diagnostic threshold, the patient is classified as test-positive, otherwise as test-negative.

### 5.1. Version

1.0.6

### 5.2. ODA

[10.5281/zenodo.20209103](https://zenodo.org/record/20209103)

### 5.3. Source Code

*Programming language:* Wolfram Language

*Software source code file format:* Wolfram Notebook

*Availability:* The updated source code is available at: <https://www.hcsl.com/Tools/Relation/Relation.nb>

### 5.4. Software Requirements

*Operating systems:* Microsoft Windows, Linux, Apple macOS and iOS

*Other software requirements:* Wolfram Player®, freely available at: <https://www.wolfram.com/player/> or Wolfram Mathematica®.



## 5.5. System Requirements

*Processor:* Intel Core i7® or equivalent CPU

*System memory (RAM):* 16 GB+ recommended.

## 6. Citation

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